

Doctor Dashboard - Functional Requirements Document

1. Document Overview

1.1 Purpose

This document defines the functional requirements for the Doctor Dashboard within the Sarthi Platform, enabling physicians to conduct AI-assisted clinical encounters with real-time documentation, billing code generation, and EMR integration.

1.2 Scope

The Doctor Dashboard provides pre-visit intelligence, live AI scribe functionality, real-time clinical decision support, and seamless integration with EMR systems and other Sarthi portals.

1.3 Stakeholders

- **Primary Users:** Physicians, Nurse Practitioners, Physician Assistants
 - **Secondary Users:** Medical Assistants (for assistance requests), Billing Staff (for generated claims)
 - **System Integrators:** EMR vendors, Laboratory systems, Insurance providers
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2. Functional Requirements

FR-DD-001: Dashboard Display and Navigation

Requirement: The system SHALL provide a role-based dashboard for physicians showing patient queues, schedules, and pending lab results.

User Story: As a physician, I want to see all patients requiring my attention in organized queues so that I can prioritize my workflow efficiently.

Acceptance Criteria:

- AC-DD-001.1: Dashboard MUST display "Ready for You" section showing patients roomed by MA
- AC-DD-001.2: Dashboard MUST display "Today's Full Schedule" with all scheduled patients and their status
- AC-DD-001.3: Dashboard MUST display "Lab Results for Final Review" requiring physician approval
- AC-DD-001.4: Dashboard MUST refresh automatically every 30 seconds
- AC-DD-001.5: Dashboard MUST load within 2 seconds on modern laptop hardware
- AC-DD-001.6: Dashboard MUST display EMR sync status and last sync time

- AC-DD-001.7: Dashboard MUST be responsive and usable on laptop screens (1366x768 minimum)

Priority: High

FR-DD-002: Pre-Visit Intelligence System

Requirement: The system SHALL automatically generate AI-powered clinical analysis for upcoming appointments using a configurable 7-14 day look-ahead window.

User Story: As a physician, I want AI-generated patient analysis and suggested care plans before each visit so that I can provide more efficient and comprehensive care.

Acceptance Criteria:

- AC-DD-002.1: System MUST pull patient data from EMR, lab systems, insurance APIs, and scheduling systems
- AC-DD-002.2: System MUST process data using rolling 4-hour intervals to distribute computational load
- AC-DD-002.3: System MUST treat EMR as single source of truth for all data conflicts
- AC-DD-002.4: System MUST generate clinical summary within 30 seconds of data changes
- AC-DD-002.5: System MUST provide suggested checklist based on patient history and visit type
- AC-DD-002.6: System MUST identify clinical alerts (allergies, drug interactions, critical values)
- AC-DD-002.7: System MUST cache analysis results for offline access
- AC-DD-002.8: System MUST purge analysis data immediately upon appointment cancellation
- AC-DD-002.9: System MUST preserve analysis data for rescheduled appointments

Priority: High

FR-DD-003: Patient Chart Access and Navigation

Requirement: The system SHALL provide comprehensive patient chart interface with AI-powered analysis, vital signs, medication review, and visit history.

User Story: As a physician, I want to access complete patient information in a single interface so that I can make informed clinical decisions quickly.

Acceptance Criteria:

- AC-DD-003.1: Chart MUST display patient demographics, visit reason, and room assignment

- AC-DD-003.2: Chart MUST show AI-generated analysis and suggested checklist
- AC-DD-003.3: Chart MUST display current vital signs entered by MA
- AC-DD-003.4: Chart MUST separate practice-prescribed medications from patient-reported medications
- AC-DD-003.5: Chart MUST show medication adherence scoring when available
- AC-DD-003.6: Chart MUST display past visits and vital sign trends
- AC-DD-003.7: Chart MUST provide "Request MA Assistance" functionality
- AC-DD-003.8: Chart MUST show real-time EMR sync status
- AC-DD-003.9: Chart MUST be accessible via "View Chart" button from dashboard

Priority: High

FR-DD-004: Live AI Scribe System

Requirement: The system SHALL provide real-time audio transcription with AI-powered clinical documentation during patient encounters.

User Story: As a physician, I want to focus on patient care while AI automatically generates clinical documentation so that I can maintain eye contact and provide better patient experience.

Acceptance Criteria:

- AC-DD-004.1: System MUST provide live audio transcription as primary mode
- AC-DD-004.2: System MUST monitor bandwidth and fallback to record-and-transcribe if <2 Mbps or >500ms latency
- AC-DD-004.3: System MUST display persistent banner notification for connectivity issues
- AC-DD-004.4: System MUST cache encounter data during offline mode and sync when connectivity restored
- AC-DD-004.5: System MUST provide visual waveform display during recording
- AC-DD-004.6: System MUST identify speaker (Doctor vs Patient) in transcription
- AC-DD-004.7: System MUST highlight medical terminology and provide confidence scoring
- AC-DD-004.8: System MUST allow manual pause/resume of recording
- AC-DD-004.9: System MUST provide "mark important" functionality for key moments
- AC-DD-004.10: System MUST allow private notes that are not transcribed

Priority: High

FR-DD-005: Multi-Agent AI Coordination

Requirement: The system SHALL coordinate multiple AI agents simultaneously during live encounters to generate clinical documentation, billing codes, and orders.

User Story: As a physician, I want multiple AI assistants working simultaneously during my patient encounter so that all aspects of documentation are handled efficiently.

Acceptance Criteria:

- AC-DD-005.1: System MUST run 5 AI agents simultaneously: Clinical Scribe, Billing Coder, Order Manager, Medication Reconciler, Clinical Decision Support
- AC-DD-005.2: System MUST process all AI responses within 5 seconds
- AC-DD-005.3: System MUST continue operation if one agent fails and flag manual completion needs
- AC-DD-005.4: System MUST allow physician to choose between conflicting AI suggestions
- AC-DD-005.5: System MUST prioritize Clinical Decision Support agent for safety alerts (<2 seconds)
- AC-DD-005.6: System MUST use GCP auto-scaling for AI processing resources
- AC-DD-005.7: System MUST log all AI suggestions and physician decisions for training data

Priority: High

FR-DD-006: SOAP Note Generation

Requirement: The system SHALL generate structured SOAP notes in real-time using practice-specific templates and medical terminology.

User Story: As a physician, I want AI to generate properly formatted SOAP notes using my practice's templates and terminology so that documentation meets my standards and EMR requirements.

Acceptance Criteria:

- AC-DD-006.1: System MUST select templates based on visit type, provider preference, and patient condition
- AC-DD-006.2: System MUST use practice-customizable medical shorthand and terminology
- AC-DD-006.3: System MUST populate SOAP sections in real-time during encounter
- AC-DD-006.4: System MUST incorporate EMR-specific formatting and required fields
- AC-DD-006.5: System MUST allow live editing of generated notes by physician

- AC-DD-006.6: System MUST auto-save note changes every 60 seconds
- AC-DD-006.7: System MUST maintain version control of AI vs physician modifications
- AC-DD-006.8: System MUST map SOAP sections to specific EMR fields per practice configuration
- AC-DD-006.9: System MUST validate note completeness before allowing signature

Priority: High

FR-DD-007: Billing Code Management

Requirement: The system SHALL suggest appropriate billing codes dynamically during the encounter and allow physician review and modification.

User Story: As a physician, I want AI to suggest appropriate billing codes during my encounter so that I can ensure proper reimbursement without disrupting patient care.

Acceptance Criteria:

- AC-DD-007.1: System MUST suggest billing codes based on pre-analysis and encounter content
- AC-DD-007.2: System MUST display codes with priority ordering based on confidence scores
- AC-DD-007.3: System MUST update code suggestions dynamically as encounter evolves
- AC-DD-007.4: System MUST validate medical necessity alignment between diagnoses and procedures
- AC-DD-007.5: System MUST check insurance coverage for suggested procedures
- AC-DD-007.6: System MUST detect appropriate modifiers for procedures
- AC-DD-007.7: System MUST allow physician to accept, modify, reject, or add custom codes
- AC-DD-007.8: System MUST provide bulk actions for multiple code approval
- AC-DD-007.9: System MUST warn if documentation is insufficient for selected codes

Priority: High

FR-DD-008: Order Management and Prior Authorization

Requirement: The system SHALL suggest clinical orders and automatically detect prior authorization requirements.

User Story: As a physician, I want AI to suggest appropriate orders and alert me to prior authorization requirements so that I can streamline the ordering process and avoid delays.

Acceptance Criteria:

- AC-DD-008.1: System MUST suggest laboratory, imaging, prescription, and referral orders based on encounter
- AC-DD-008.2: System MUST check real-time insurance databases for prior authorization requirements
- AC-DD-008.3: System MUST flag orders requiring prior authorization with "Request Prior Auth" status
- AC-DD-008.4: System MUST suggest covered alternatives when prior auth is required
- AC-DD-008.5: System MUST perform real-time drug interaction checking against current medications
- AC-DD-008.6: System MUST cross-reference orders against patient allergies
- AC-DD-008.7: System MUST check drug formulary preferences for insurance plans
- AC-DD-008.8: System MUST allow physician to approve, modify, or reject suggested orders
- AC-DD-008.9: System MUST transmit approved orders via API to appropriate systems

Priority: High

FR-DD-009: Split-Screen Mode and Voice Commands

Requirement: The system SHALL provide split-screen mode with voice-activated display of patient test results and clinical data.

User Story: As a physician, I want to view relevant test results using voice commands during the encounter so that I can reference data without interrupting patient conversation.

Acceptance Criteria:

- AC-DD-009.1: System MUST prefetch all historical lab results, imaging, and consultation notes during pre-analysis
- AC-DD-009.2: System MUST activate split-screen mode via button click or voice command
- AC-DD-009.3: System MUST provide 60/40 split layout (scribe left, results right) on desktop
- AC-DD-009.4: System MUST respond to semantic voice commands within 2 seconds
- AC-DD-009.5: System MUST support commands like "Show latest CBC", "Display A1C trend", "Pull up liver function"
- AC-DD-009.6: System MUST present multiple matching results for physician selection
- AC-DD-009.7: System MUST provide numbered shortcuts (1,2,3) for quick selection
- AC-DD-009.8: System MUST remember frequently accessed results for quick access

- AC-DD-009.9: System MUST work seamlessly on laptop screen sizes

Priority: Medium

FR-DD-010: Post-Visit Care Note Generation

Requirement: The system SHALL automatically generate patient-friendly care notes after each encounter with multi-language support and personalization.

User Story: As a physician, I want AI to generate clear, personalized care instructions for my patients so that they understand their care plan and follow through effectively.

Acceptance Criteria:

- AC-DD-010.1: System MUST generate care notes personalized by condition, visit type, and patient demographics
- AC-DD-010.2: System MUST use simple language (grade 6 reading level maximum)
- AC-DD-010.3: System MUST support multiple languages (English, Spanish, Chinese, Vietnamese, Arabic, Russian)
- AC-DD-010.4: System MUST include visit summary, medication changes, next steps, and warning signs
- AC-DD-010.5: System MUST deliver via patient portal, email, SMS, or print
- AC-DD-010.6: System MUST adapt content for different age groups (pediatric, adult, geriatric)
- AC-DD-010.7: System MUST match care note content to encounter transcript accuracy
- AC-DD-010.8: System MUST allow optional physician review before patient delivery
- AC-DD-010.9: System MUST track patient receipt and reading confirmation

Priority: Medium

FR-DD-011: Lab Result Review and Approval

Requirement: The system SHALL provide streamlined lab result review with AI interpretation and approval workflow.

User Story: As a physician, I want to efficiently review and approve lab results with AI assistance so that patients receive timely results and appropriate follow-up care.

Acceptance Criteria:

- AC-DD-011.1: System MUST display pending lab results requiring physician approval
- AC-DD-011.2: System MUST provide AI-generated preliminary interpretation for review
- AC-DD-011.3: System MUST open detailed review dialog when "Review Results" is clicked
- AC-DD-011.4: System MUST pre-fill physician notes with AI interpretation
- AC-DD-011.5: System MUST provide "Accept AI Suggestion" and "Fix Manually" options
- AC-DD-011.6: System MUST disable approval until physician makes selection
- AC-DD-011.7: System MUST allow "Send Back to Technician" with specific feedback
- AC-DD-011.8: System MUST update result status and notify appropriate portals upon approval
- AC-DD-011.9: System MUST automatically release approved results to patient portal and EMR

Priority: High

FR-DD-012: Real-Time State Management

Requirement: The system SHALL implement patient-level atomic operations with real-time synchronization across all portals.

User Story: As a physician, I want exclusive access to patient records during encounters while allowing MA assistance requests so that data integrity is maintained across the system.

Acceptance Criteria:

- AC-DD-012.1: System MUST implement patient-level check-in/check-out locking mechanism
- AC-DD-012.2: System MUST provide exclusive access to physician during "encounter_in_progress" state
- AC-DD-012.3: System MUST allow MA assistance requests as only exception during encounter
- AC-DD-012.4: System MUST broadcast status changes to all portals within 1 second
- AC-DD-012.5: System MUST use WebSocket connections with polling fallback for real-time updates
- AC-DD-012.6: System MUST prevent concurrent access conflicts through atomic operations
- AC-DD-012.7: System MUST notify other users when patient record is locked
- AC-DD-012.8: System MUST release locks automatically upon encounter completion
- AC-DD-012.9: System MUST maintain audit trail of all access attempts and lock changes

Priority: High

FR-DD-013: EMR Integration and Synchronization

Requirement: The system SHALL synchronize all encounter data with EMR systems post-encounter with manual fallback options.

User Story: As a physician, I want all my documentation to sync automatically with our EMR so that patient records are complete and billing can be processed properly.

Acceptance Criteria:

- AC-DD-013.1: System MUST sync encounter data to EMR upon encounter completion
- AC-DD-013.2: System MUST map SOAP notes to specific EMR fields per practice configuration
- AC-DD-013.3: System MUST validate data before synchronization and report errors
- AC-DD-013.4: System MUST provide manual sync button for technical issues
- AC-DD-013.5: System MUST offer structured data export if automatic sync fails
- AC-DD-013.6: System MUST maintain complete audit trail of all sync attempts
- AC-DD-013.7: System MUST allow rollback of failed synchronizations
- AC-DD-013.8: System MUST queue failed syncs for retry with exponential backoff
- AC-DD-013.9: System MUST alert physician of sync failures and provide resolution options

Priority: High

FR-DD-014: Clinical Safety and Decision Support

Requirement: The system SHALL provide comprehensive clinical decision support with human-in-the-loop safety mechanisms.

User Story: As a physician, I want immediate alerts for drug interactions, allergies, and critical values so that I can provide safe patient care with AI assistance.

Acceptance Criteria:

- AC-DD-014.1: System MUST require physician review and approval for all AI suggestions
- AC-DD-014.2: System MUST display drug interactions with red lettering and modal cards
- AC-DD-014.3: System MUST integrate with clinical databases (Lexicomp, Micromedex, UpToDate)
- AC-DD-014.4: System MUST cross-reference all medications against known allergies
- AC-DD-014.5: System MUST flag critical lab values with immediate notifications
- AC-DD-014.6: System MUST require acknowledgment before proceeding with critical alerts

- AC-DD-014.7: System MUST provide clinical guideline reminders and evidence links
- AC-DD-014.8: System MUST suggest safer alternatives for contraindicated medications
- AC-DD-014.9: System MUST log all safety alerts and physician responses

Priority: Critical

FR-DD-015: Performance and Scalability

Requirement: The system SHALL support 500+ concurrent providers across multiple practices with enterprise-grade performance.

User Story: As a system administrator, I want the platform to handle our entire multi-practice organization efficiently so that physicians experience consistent performance regardless of system load.

Acceptance Criteria:

- AC-DD-015.1: System MUST support minimum 500 concurrent providers across multiple practices
- AC-DD-015.2: System MUST use GCP auto-scaling infrastructure with no artificial resource limits
- AC-DD-015.3: System MUST prioritize patient care workloads over administrative functions
- AC-DD-015.4: System MUST require modern laptop minimum specifications (8GB RAM, i5 processor, 25Mbps network)
- AC-DD-015.5: System MUST load dashboard within 2 seconds (95th percentile)
- AC-DD-015.6: System MUST complete AI processing within 5 seconds for all agents
- AC-DD-015.7: System MUST sync EMR data within 10 seconds post-encounter
- AC-DD-015.8: System MUST propagate real-time updates across portals within 1 second
- AC-DD-015.9: System MUST maintain 99.9% uptime during business hours

Priority: High

FR-DD-016: Integration with External Systems

Requirement: The system SHALL integrate with ePrescribing, laboratory, and insurance systems through existing EMR capabilities and API connections.

User Story: As a physician, I want seamless integration with all our existing systems so that I don't have to duplicate work or learn new workflows.

Acceptance Criteria:

- AC-DD-016.1: System MUST leverage existing EMR ePrescribing capabilities rather than creating new integrations
- AC-DD-016.2: System MUST sync medication data to EMR which triggers ePrescribing workflow
- AC-DD-016.3: System MUST integrate with laboratory systems via API for order transmission and result retrieval
- AC-DD-016.4: System MUST perform real-time insurance verification via API (Availity, Change Healthcare)
- AC-DD-016.5: System MUST detect prior authorization requirements automatically
- AC-DD-016.6: System MUST calculate patient copays and deductibles in real-time
- AC-DD-016.7: System MUST provide fallback to manual processes if integrations fail
- AC-DD-016.8: System MUST support multiple EMR systems with configurable field mapping
- AC-DD-016.9: System MUST maintain integration monitoring and error reporting

Priority: High

3. Non-Functional Requirements

NFR-DD-001: Security and Compliance

- System MUST comply with HIPAA regulations for PHI handling
- System MUST encrypt all data in transit (TLS 1.3) and at rest (AES-256)
- System MUST log all patient access and data modifications for audit compliance
- System MUST implement role-based access control with session timeouts

NFR-DD-002: Usability

- System MUST be accessible via keyboard navigation for disability compliance
- System MUST support screen readers and high contrast mode
- System MUST provide tooltips and help text for all complex features
- System MUST maintain consistent UI patterns across all interfaces

NFR-DD-003: Reliability

- System MUST have 99.9% uptime during business hours
- System MUST gracefully handle all error conditions without data loss
- System MUST provide automatic backup and recovery capabilities

- System MUST support disaster recovery with <4 hour RTO

NFR-DD-004: Maintainability

- System MUST use modular architecture for easy updates and feature additions
 - System MUST provide comprehensive logging for troubleshooting
 - System MUST support A/B testing for UI and workflow improvements
 - System MUST allow configuration changes without system downtime
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4. Constraints and Assumptions

Technical Constraints

- Must run on modern laptop hardware (minimum specifications defined)
- Must integrate with existing EMR systems without replacing core functionality
- Must use GCP infrastructure for AI processing and data storage
- Must support Chrome, Firefox, and Safari browsers (latest versions)

Business Constraints

- Must not reinvent existing EMR capabilities (ePrescribing, core medical records)
- Must maintain physician workflow efficiency without adding administrative burden
- Must provide immediate ROI through reduced documentation time
- Must scale to support multi-practice deployments

Regulatory Constraints

- Must comply with HIPAA privacy and security requirements
 - Must support medical billing and coding compliance standards
 - Must maintain complete audit trails for regulatory compliance
 - Must support data export for regulatory reporting requirements
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5. Acceptance Criteria Summary

The Doctor Dashboard system will be considered complete when:

1. All functional requirements (FR-DD-001 through FR-DD-016) are implemented and tested
2. All acceptance criteria are met and verified through user acceptance testing

3. Performance benchmarks are achieved under simulated production load
4. Security and compliance requirements are validated by third-party audit
5. Integration testing with EMR, lab, and insurance systems is successful
6. Multi-practice deployment testing demonstrates scalability requirements